Geography 7 — Solved Exercises

Class 7 · Punjab Curriculum and Textbook Board (PCTB)

Solved Exercises

COMPLETE BOOK: all 6 chapters — chapter-end Exercises (Q1 MCQs, Q2 short, Q3 detailed), Critical Thinking Questions and Projects for every chapter, plus ALL in-text Skills boxes in every chapter (Ch 1: pp. 2, 4, 8; Ch 2: pp. 18, 22, 25; Ch 3: p. 39; Ch 4: pp. 51, 53; Ch 5: pp. 61, 63, 64; Ch 6: pp. 80, 84). Geography SNC, 93 book pages. Every content page of the book (pp. 1-87) was paged through in the follow-up pass; glossary/bibliography (pp. 88-90) carry no exercises.

Chapter 1 — Landforms: Plains

Exercise — Skills (Chapter 1, book p. 2)

Q1. Skills: Investigate reasons why the Arctic Plain is known as a barren plain?

Answer: The Arctic Plain is called a barren plain because its ground stays frozen for most of the year, so trees and tall plants cannot grow there.

The word Tundra itself comes from the Finnish word “tunturi”, meaning a treeless plain, and the chapter describes Arctic plains as frozen, treeless land.

With almost no trees, very little plant growth and long, cold winters, the land looks bare and empty, which is why it is called barren.

Q2. Skills: Investigate the living conditions in a desert or in the Arctic Tundra Plain.

Answer: Life in a desert or in the Arctic Tundra Plain is hard because of extreme weather and very little water or plant growth.

In a desert, days are very hot and dry, rain is scarce, and only plants and animals that can survive with little water live there.

In the Arctic Tundra, the ground is frozen, trees cannot grow, and people and animals must bear long, bitterly cold winters with only short, cool summers.

In both places people depend on special clothing, shelter and food storage, and they keep animals suited to the climate, such as camels in deserts and reindeer in the far north.

Skills — Chapter 1 (book p. 4)

Skills: Search images and photographs of plains and paste them in your notebooks.

Answer: Collect labelled pictures of the main kinds of plains and paste them in your notebook under their names.

Use a river plain (for example the Indus Plain), a grassland plain (steppes), a coastal plain and the frozen tundra plain.

Under each picture write one line saying how that plain was formed and where it is found.

Skills — Chapter 1 (book p. 8)

Skills: Inquire about the working of a river system.

Answer: A river system is a main river together with all its tributaries and the land they drain.

In the upper course the river erodes steep valleys; in the middle course it meanders and deposits silt on flood plains.

In the lower course it deposits its load and forms a delta, and the whole system carries water and sediment to the sea.

Exercise — Chapter 1 (book pp. 12-13)

Q1 (i). In Asia and Eastern Europe, temperate grasslands are called:

(a) tundra

(b) steppes ✓

(c) deserts

(d) tropical

Answer: (b) steppes — the chapter states that in Asia and Eastern Europe

temperate grasslands are called steppes.

Q1 (ii). Plains are vast stretches of:

(a) flat land ✓

(b) frozen land

(c) rocky land

(d) sandy land

Answer: (a) flat land — a plain is a broad area of relatively flat land.

Q1 (iii). The highest natural landscapes are:

(a) plains

(b) plateaus

(c) mountains ✓

(d) valleys

Answer: (c) mountains — the chapter's summary states mountains are the

highest natural landscapes.

Q1 (iv). The Kashmir Valley is an example of:

(a) Alluvial Plain

(b) Flood Plain

(c) Lacustrine Plain ✓

(d) Wind Plain

Answer: (c) Lacustrine Plain — a plain formed by deposits in a former lake basin, as in the Kashmir Valley.

Q1 (v). Repeated flooding of the river form:

(a) Meanders

(b) Levees ✓

(c) Delta

(d) Ox-bow Lake

Answer: (b) Levees — the chapter's summary states levee features are formed by the repeated flooding of the river.

Q2 (i). Define the term plain.

Answer: A plain is a broad area of relatively flat land.

Plains are one of the major landforms of the Earth and cover more than one-third of the world's land area.

They exist on every continent.

Q2 (ii). What are Plateaus?

Answer: A plateau is an area of high, flat land that rises sharply above the surrounding land.

It has a flat top and steep sides, and stands higher than a plain but lower and flatter than a mountain range.

Q2 (iii). Write the three features of Middle Course River.

Answer: The three main features of the middle course of a river are meanders, flood plains and ox-bow lakes.

In the middle course the river flows more slowly over gentler land, bends into wide curves (meanders), and deposits silt.

When a meander neck is cut off, the abandoned bend remains as an ox-bow lake beside the river.

Q2 (iv). Define the term valley.

Answer: A valley is a low area of land between hills or mountains, often with a river flowing through it.

Valleys are usually formed by the erosion of rivers or glaciers cutting down through higher land.

Q2 (v). Write the formation of flood plain.

Answer: A flood plain is formed when a river floods its banks again and again over a long period of time.

Each flood leaves behind a layer of fine silt and alluvium on the flat land beside the river.

These repeated deposits build up a wide, flat and very fertile plain.

Q3 (i). What are the physical features of plains? Describe.

Answer: Plains are broad, flat or gently rolling areas of low relief with few hills.

Their soil is usually deep and fertile, especially where rivers have deposited alluvium, so they suit farming.

Many plains are grasslands, and the Indus Plain and the Great Plains are large examples given in the chapter.

Q3 (ii). Explain some major types of plains.

Answer: Alluvial plains are built by river deposits, and flood plains by repeated river flooding.

Coastal plains lie along sea coasts, and lacustrine plains form in former lake basins, like the Kashmir Valley.

Erosional plains are worn flat by rivers, glaciers and wind, while the frozen Arctic plains are called tundra.

Q3 (iii). Differentiate plains from the landforms like plateaus, mountains and valley, etc.

Answer: Plains are low, flat and broad, while plateaus are high flat-topped areas raised above the surrounding land.

Mountains are the highest landforms, steep and rugged, whereas a valley is a low depression between hills or mountains.

In short: plains are low and flat, plateaus are high and flat, mountains are high and steep, and valleys are low and enclosed.

Q3 (iv). Describe the features formed by river.

Answer: In its upper course a river erodes steep valleys and gorges; in the middle course it forms meanders and flood plains.

Repeated flooding builds levees along its banks, and cut-off meanders become ox-bow lakes.

In the lower course the river deposits its load and forms a delta where it meets the sea or a lake.

Q3 (v). What is a river system? Explain.

Answer: A river system is a main river together with all its tributaries and the area they drain.

Small streams join to form larger tributaries, and all of them flow into the main river.

The whole system drains one basin, carrying water and sediment from the surrounding land to the sea or a lake.

Q3 (vi). Compare the middle course of river with lower course of river.

Answer: In the middle course the land is gentler, the river flows more slowly, meanders widely and begins depositing silt.

In the lower course the land is almost flat, the river is wide and slow, and deposition dominates over erosion.

The middle course forms meanders, flood plains and ox-bow lakes, while the lower course forms levees and deltas.

Q3 (vii). Explore the importance of rivers in the economy of a country.

Answer: Rivers irrigate crops, supply drinking water and support fishing, which sustain farming and food production.

They provide cheap transport, generate hydropower and recharge groundwater used by towns and farms.

River plains attract settlement, industry and tourism, so rivers raise incomes, trade and the overall economy.

Critical Thinking Questions — Chapter 1 (book p. 14)

Critical Thinking: The relief of Pakistan displays a great physical variation. Explain how?

Answer: Pakistan's relief varies from the high Himalaya, Karakoram and Hindu Kush mountains in the north to the Potohar Plateau.

Below them lies the vast, flat Indus Plain, while Balochistan has its own plateau and dry mountains.

In the east and south-east are deserts, and in the south the Indus delta meets the Arabian Sea coast.

Critical Thinking: Why are river plains thickly populated?

Answer: River plains are thickly populated because their flat, fertile soil gives rich harvests and water is easily available.

Flat land makes building houses, roads and towns easy, and rivers provide transport, fishing and trade.

Jobs in farming, markets and industry therefore gather people on river plains.

Critical Thinking: Which two factors can change a river?

Answer: The two main factors that change a river are erosion and deposition, which alter its course, depth and shape.

Human actions also change rivers: dams store water, embankments control floods, and irrigation canals divert flow.

Together, natural processes and human works can shift a river's channel and change the land around it.

Critical Thinking: How do dams affect the river and its environment?

Answer: A dam stores river water, reducing flow downstream and trapping silt that once fertilised flood plains.

This changes habitats for fish and wildlife and can lower water for downstream farms and wetlands.

At the same time dams supply irrigation water, hydropower and flood control, so their effects are both useful and harmful.

Critical Thinking: Why do deltas attract people?

Answer: Deltas attract people because the river deposits rich, fertile silt that suits intensive farming.

Fishing, ports and trade grow at deltas, giving jobs and easy transport by water.

Flat land, water and markets together make deltas busy and densely settled places.

Project For Students — Chapter 1 (book p. 14)

Project: Make a model of a River System, label the different features of river and share with your classmates.

Answer: Project task (to be made, not written): build a model showing the upper, middle and lower courses of a river.

Label the source, tributaries, valley, meanders, flood plain, levee, ox-bow lake and delta on the model.

Present the model to the class and explain how the river changes from its source to its mouth.

Chapter 2 — Water Sources and Management

Skills — Chapter 2 (book p. 18)

Skills: Draw and label Water Cycle.

Answer: Draw the water cycle as a circle: evaporation from seas, lakes and land, then condensation into clouds.

Label precipitation (rain, snow, hail), run-off over the ground, infiltration into the soil, and transpiration from plants.

Mark the four spheres it links: atmosphere, lithosphere, hydrosphere and biosphere.

Skills: Devise ways to solve the shortage of water in Pakistan.

Answer: Store more water by building dams and small reservoirs, and repair leaking pipes so less water is wasted.

Use water wisely in farming with drip irrigation and lined canals, and harvest rainwater in homes and schools.

Treat and reuse wastewater for irrigation, and teach people to save water instead of wasting it.

Skills — Chapter 2 (book p. 22)

Skills: Evaluate ways of using wastewater.

Answer: Treated sewage and grey water can be used for watering crops, parks and gardens, and for industry.

It can also recharge groundwater and clean streets, which saves fresh water for drinking.

Wastewater must be treated first: untreated sewage spreads disease, so it is never used directly for drinking.

Skills — Chapter 2 (book p. 25)

Skills: Organize information about water purification methods into flow charts.

Answer: Flow chart of the four water purification methods in the chapter: Boiling, then Filtration, then Chlorination, then Reverse Osmosis.

Boiling: heat water to a rolling boil for 1-3 minutes to kill germs and parasites.

Filtration: pass water through chemical and physical filters to remove sediment and chlorine.

Chlorination: add chlorine tablets to heated water to kill germs; Reverse Osmosis: force water through a fine membrane that removes dissolved impurities.

Skills: Analyze the reasons for the scarcity of drinking water in cities.

Answer: Cities face drinking-water scarcity because their growing population raises demand faster than supply.

Leaking pipes, wastage and poor management lose much water before it reaches homes.

Pollution of rivers and falling groundwater levels further reduce the clean water available to cities.

Exercise — Chapter 2 (book p. 26)

Q1 (i). The cheapest naturally occurring water source:

(a) river

(b) glacier

(c) ocean

(d) rainwater ✓

Answer: (d) rainwater — rain falls freely and needs no pumping or treatment before collection, so it is the cheapest natural source.

Q1 (ii). The Water Cycle begins with:

(a) evaporation ✓

(b) condensation

(a) transpiration

(d) infiltration

Answer: (a) evaporation — heat from the Sun evaporates water from seas, lakes and land, starting the cycle.

Q1 (iii). Water covers of the Earth's surface:

(a) about 69%

(b) about 71% ✓

(c) about 73%

(d) about 75%

Answer: (b) about 71% — the chapter's summary states water covers about 71% of the Earth's surface.

Q1 (iv). Water purification methods are:

(a) 2

(b) 3

(c) 4 ✓

(d) 5

Answer: (c) 4 — the chapter states there are four water purification methods: boiling, filtration, chlorination and reverse osmosis.

Q1 (v). The river is estimated to carry 20% of all the fresh water found on the Earth:

(a) Indus

(b) Amazon ✓

(c) Nile

(d) Tigris

Answer: (b) Amazon — the Amazon carries about one-fifth of all the fresh water discharged by the world's rivers.

Q2 (i). What is orographic rainfall?

Answer: Orographic rainfall occurs when moist air is forced to rise over a mountain barrier.

As the air rises it cools, water vapour condenses into clouds, and rain falls on the windward side of the mountain.

The far side, in the rain shadow, receives much less rain.

Q2 (ii). Write two uses of biogas.

Answer: Biogas is used as a cooking and heating fuel in homes, replacing wood and fossil fuels.

It is also used to generate electricity and as a vehicle fuel after upgrading to biomethane.

Q2 (iii). Define the term Chlorination.

Answer: Chlorination is the method of purifying water by adding chlorine, usually as chlorine tablets, to kill germs.

The chapter notes that chlorine tablets should be applied in heated water.

It makes water safe to drink by destroying disease-causing microbes.

Q2 (iv). What are aquifers?

Answer: Aquifers are underground layers of rock, sand or gravel that hold groundwater.

Water collects in their pores and cracks below the surface of the Earth.

Wells and tube wells draw drinking and irrigation water from aquifers.

Q2 (v). Which instrument is used to measure rain?

Answer: Rain is measured with a rain gauge.

A rain gauge is an open container that collects rainfall, and the depth of collected water shows how much rain fell.

It is usually read in millimetres over a fixed period, such as a day.

Q3 (i). Explain the major sources of water on Earth and Pakistan? Explain.

Answer: On Earth the major sources are oceans, glaciers, lakes, rivers, rain and groundwater stored in aquifers.

Oceans are the largest source, while glaciers store most of the world's fresh water as ice.

In Pakistan the main sources are the Indus River and its tributaries, Himalayan glaciers, rainfall, dams and groundwater.

Q3 (ii). Describe the different types of precipitation.

Answer: Precipitation is water falling from clouds in liquid or solid form.

Rain is liquid drops; drizzle is fine, light rain; snow is soft ice crystals falling in cold weather.

Sleet is partly frozen rain, and hail is hard balls of ice formed in strong thunderstorm clouds.

Q3 (iii). Explain the process of Water Cycle in maintaining water supply of the Earth.

Answer: The Sun's heat evaporates water from oceans, lakes and land, and plants add water by transpiration.

Water vapour cools and condenses into clouds, then falls back as precipitation over land and sea.

Part of it flows in rivers, part sinks into aquifers, and the cycle repeats, constantly renewing the Earth's fresh water supply.

Q3 (iv). Describe different uses of sewage waste.

Answer: Treated sewage waste is used to produce biogas, a renewable fuel for cooking, heating and electricity.

Its sludge, after treatment, can be used as manure that adds nutrients to farm soil.

Treated wastewater can also be reused for irrigation and industrial purposes, saving fresh water.

Q3 (v). Discuss the biogas as a source of energy.

Answer: Biogas is a renewable fuel produced by breaking down organic matter such as food scraps and animal waste without oxygen.

It can be used for cooking, heating, electricity generation and, upgraded to biomethane, as vehicle fuel.

It is a greener fuel because it replaces fossil fuels and uses waste that would otherwise pollute.

Critical Thinking Questions — Chapter 2 (book p. 27)

Critical Thinking: What will happen if there is no Water Cycle?

Answer: Without the water cycle there would be no rain to refill rivers, lakes and aquifers.

Fresh water supplies would run out, crops would fail, and plants, animals and people would face severe drought.

Life on Earth depends on the cycle constantly moving water between sea, sky and land.

Critical Thinking: Can biogas reduce climate change?

Answer: Yes, biogas can reduce climate change because it replaces coal, oil and gas, cutting carbon dioxide emissions.

It also captures methane from rotting waste, a powerful greenhouse gas, and burns it as useful fuel.

Using waste for energy therefore lowers both fossil-fuel use and uncontrolled methane release.

Critical Thinking: How does sewage affect water quality and aquatic life?

Answer: Untreated sewage adds germs, chemicals and nutrients to rivers and lakes, making water unsafe to drink.

Excess nutrients cause algal growth that uses up oxygen, so fish and other aquatic life suffocate and die.

Sewage pollution also spreads disease and damages the whole river ecosystem.

Critical Thinking: Which regions received the highest rate of precipitation and why?

Answer: The highest precipitation falls in equatorial regions and on windward mountain slopes and monsoon coasts.

Near the equator, warm moist air rises strongly, cools and condenses into heavy rain all year.

Where moist winds are forced up mountains, orographic rain adds further heavy rainfall.

Project For Students — Chapter 2 (book p. 27)

Project: Make a project on conservation of water purification methods.

Answer: Project task (to be made, not written): prepare a project showing the four purification methods — boiling, filtration, chlorination and reverse osmosis.

Include a flow chart for each method, its merits and demerits, and tips for conserving purified water at home.

Present the project and explain which method suits different situations.

Chapter 3 — Settlements and Land Use

Skills — Chapter 3 (book p. 39)

Skills: Investigate how the pattern of a settlement determines the lifestyle of the people living in it.

Answer: In a nucleated settlement houses stand close together, so people share schools, markets and services and live a close community life.

In a dispersed settlement houses stand far apart, so families depend more on their own farms and travel further for services.

In a linear settlement along a road or river, trade and transport shape daily life, while a planned grid settlement gives orderly streets and services.

Skills: Predict that there will be no land left on Earth if settlements will keep growing at the current pace.

Answer: If towns and cities keep spreading at the present pace, houses, roads and factories will cover more and more open land.

Farmland, forests and natural habitats will shrink, leaving less land for growing food and for wildlife.

To avoid this, growth must be planned: build upwards in cities, protect farmland, and reuse land already built on.

Critical Thinking Questions — Chapter 3 (book p. 40)

Critical Thinking: How did human life styles change with settlements?

Answer: Settled life replaced wandering, as people began farming, storing food and living in permanent villages.

Villages grew into towns and cities with markets, schools, hospitals and different jobs beyond farming.

Lifestyles changed from self-sufficient farming to specialised work, trade and urban services.

Critical Thinking: How do settlements change over time?

Answer: Settlements grow as population rises: hamlets become villages, villages become towns, and towns merge into cities.

New roads, railways, industries and services attract more people and expand the built-up area.

Over time neighbouring towns may join into conurbations, changing land use from fields to housing and industry.

Critical Thinking: How can settlements be more sustainable?

Answer: Settlements become sustainable with planned housing, clean water supply and proper waste disposal and recycling.

Public transport, parks and green spaces reduce pollution and keep the environment healthy.

Saving energy and water, and protecting farmland around towns, keeps settlements livable for future generations.

Critical Thinking: Explore the possible problems, threats that cities may face in future.

Answer: Cities may face overcrowding, traffic congestion, housing shortages and rising pollution in future.

Water scarcity, waste disposal and flooding or heatwaves from climate change threaten health and infrastructure.

Unplanned growth may also create slums and strain schools, hospitals and jobs.

Exercise — Chapter 3 (book p. 41)

Q1 (i). It is the largest type of settlement, containing lot of buildings and people:

(a) village

(b) town

(c) city ✓

(d) hamlet

Answer: (c) city — the chapter's summary states a city is the largest type of settlement, containing lots of buildings and people.

Q1 (ii). In the recent 2017 census, Pakistan's urban population is about:

(a) 30%

(b) 32%

(c) 34%

(d) 37% ✓

Answer: (d) 37% — the chapter's summary states that in the 2017 census 63% of Pakistan's population was rural and 37% was urban.

Q1 (iii). The place where people visit to relax and enjoy themselves is called:

(a) Tourist resort ✓

(b) Residential town

(c) Business centre

(d) Town market

Answer: (a) Tourist resort — a place built for visitors to relax and enjoy themselves.

Q1 (iv). By population, the biggest city of Pakistan:

(a) Lahore

(b) Karachi ✓

(c) Faisalabad

(d) Islamabad

Answer: (b) Karachi — Karachi is Pakistan's largest city by population.

Q1 (v). A place where goods can be imported and exported is called:

(a) Industrial town

(b) port ✓

(c) route centre

(d) service centre

Answer: (b) port — a port is where goods are loaded and unloaded for import and export by sea.

Q2 (i). What is meant by administrative centre?

Answer: An administrative centre is a town or city where government offices are located.

It is where officials manage the surrounding area, keep records and provide public services.

Examples include district headquarters and provincial capitals.

Q2 (ii). Define the term route centre.

Answer: A route centre is a settlement where several roads, railways or other routes meet.

Because travellers and goods pass through it, markets, workshops and services grow there.

Its importance comes from its position on transport routes rather than from farming.

Q2 (iii). What kinds of activities are held in an industrial town?

Answer: In an industrial town factories manufacture goods, workshops repair machinery, and warehouses store products.

Workers are employed in production, packaging, transport and trade of manufactured items.

Markets, banks and transport services support the industries located there.

Q2 (iv). Differentiate between Conurbations and Megalopolis.

Answer: A conurbation is a region formed when a number of towns and cities grow and merge into one continuous urban area.

A megalopolis is even larger: a chain of conurbations and metropolises linked together along a corridor.

In short, a conurbation joins neighbouring towns, while a megalopolis joins whole metropolitan regions.

Q2 (v). Define the term Settlement.

Answer: A settlement is a place where people live and build homes and other buildings.

It ranges from a tiny hamlet of a few houses to a village, town or large city.

Settlements provide shelter, work, services and community life for their people.

Q3 (i). Analyze the different types of settlements.

Answer: Rural settlements include hamlets, with just a few houses, and villages, which may also have a school, shops and a post office.

Urban settlements include towns, larger than villages with secondary schools and markets, and cities, the largest type with hospitals, universities and offices.

Beyond cities are metropolises, conurbations of merged towns, and megalopolises of linked metropolitan regions.

Q3 (ii). Explain the functions of a settlement.

Answer: Settlements provide homes and shelter, and they serve as centres for work, trade and transport.

They provide services such as schools, hospitals, markets, banks and government offices.

They also serve cultural, religious and recreational functions, bringing people together for community life.

Q3 (iii). Describe how towns are different from cities.

Answer: Towns are smaller than cities, with fewer houses, smaller markets and fewer services.

A town may have primary and secondary schools and a railway station, but a city has hospitals, universities, large offices and many industries.

Cities hold far more people and buildings, and offer a much wider range of jobs and facilities than towns.

Q3 (iv). Discuss the conditions that are suitable for a settlement.

Answer: A suitable site has a reliable water supply, flat and fertile land, and a healthy climate.

It needs safety from floods and attacks, building materials, and good transport links by road or river.

Nearby resources, farmland and opportunities for work and trade also help a settlement grow and survive.

Q3 (v). Compare two major types of settlements.

Answer: Rural settlements are small, with farming as the main work, open surroundings and few services.

Urban settlements are large and crowded, with jobs in industry, trade and services, and many facilities.

Rural life is quieter and closer to nature, while urban life offers more jobs and services but also pollution and crowding.

Q3 (vi). Describe the major types of settlements in Pakistan.

Answer: Pakistan has rural settlements: hamlets of a few houses and villages with schools, shops and post offices.

Its urban settlements include market towns, industrial towns, ports and large cities such as Karachi, Lahore, Faisalabad and Islamabad.

The 2017 census found about 63% of Pakistanis rural and 37% urban, showing both types remain important.

Q3 (vii). Highlight the advantages and disadvantages that people of rural settlement face.

Answer: Advantages of rural settlement are clean air, open natural surroundings, close community ties and a lower cost of living.

Disadvantages are fewer jobs beyond farming, and limited schools, hospitals, transport and modern facilities.

Rural people may therefore migrate to towns for education, work and better services.

Q3 (viii). Highlight the advantages and disadvantages that people of urban settlement face.

Answer: Advantages of urban settlement are many jobs, good schools, hospitals, markets, transport and entertainment.

Disadvantages are overcrowding, traffic, pollution, high living costs and pressure on water, housing and services.

Urban people enjoy more opportunities but face a busier, more expensive and more polluted life.

Project For Students — Chapter 3 (book p. 42)

Project: Make a model of any type of settlement and share with your classmates.

Answer: Project task (to be made, not written): build a model of any one settlement type, such as a village, town or city.

Show its houses, roads, markets, schools and other features, and label the settlement type.

Present the model to the class and explain why that settlement looks the way it does.

Chapter 4 — Agriculture

Skills — Chapter 4 (book p. 51)

Skills: Compare and contrast different methods of farming.

Answer: Subsistence farming grows food mainly for the farmer's own family on small fields, while commercial farming grows crops on a large scale for profit. Intensive farming gets a large output from a small area with fertilisers and irrigation, while extensive farming uses vast land with machinery. Traditional methods use old tools and animals, while modern methods use machines, better seeds and scientific techniques.

Skills: Differentiate between cash and staple crops.

Answer: Cash crops are grown to sell for profit and as industrial raw material, for example cotton, sugarcane, tobacco and rice. Staple crops are grown mainly as food for people, for example wheat, maize, millet and gram. In short: cash crops earn money and feed industry, while staple crops feed the population directly.

Skills: Use maps to locate agricultural areas and products of Pakistan.

Answer: On a map of Pakistan mark the main wheat areas in Punjab, Sindh, Khyber Pakhtunkhwa and Balochistan.

Mark cotton in Punjab and Sindh, rice in Punjab and Sindh, maize in Khyber Pakhtunkhwa and Punjab, and sugarcane in Punjab, Sindh and Khyber Pakhtunkhwa.

Use a different colour or symbol for each crop and add a key so the map can be read easily.

Skills: Analyze the factors (challenges) that affect agricultural yield.

Answer: Yield depends on water supply, soil fertility, quality seed, fertiliser and control of pests and diseases.

Weather matters too: drought, floods, frost and untimely rain can ruin a crop, while good rain and sunshine raise yield.

Modern machinery, credit, storage and fair market prices also affect how much a farmer can produce and earn.

Skills: Analyze challenges related to livestock farming in Pakistan.

Answer: Livestock farmers face shortage of fodder and clean water, especially in dry seasons, and animal diseases that spread quickly.

Veterinary care, vaccination and good shelters are often lacking, and small farmers get low prices from middlemen.

Heat, floods and poor transport add to the losses, so better feed, health care and markets are needed.

Skills — Chapter 4 (book p. 53)

Skills: Predict changes in the structure of agricultural production in the future.

Answer: In future more farming will use machines, better seeds, drip irrigation and weather information from mobile phones.

Farmers will grow more high-value crops, fruits and vegetables, and more produce will be processed and packed before sale.

Small scattered fields may join into larger, planned farms, and fewer people will work the land while output per worker rises.

Skills: Evaluate the impact of agriculture on the environment.

Answer: Agriculture feeds people and supplies raw material, but it also affects the environment in harmful ways.

Too much irrigation can cause waterlogging and salinity, chemical fertilisers and pesticides pollute soil and water, and clearing land removes forests.

Livestock adds methane gas, yet careful farming — crop rotation, organic manure and saving water — reduces these harms.

Skills: Evaluate the importance of agriculture for Pakistan in terms of Pakistan's economy.

Answer: Agriculture is vital for Pakistan's economy: it provides food for the whole population and raw material for textile and other industries.

About 37.4% of the labour force works directly in agriculture and about 70% of the people are linked to it directly or indirectly.

It contributes about one-fifth of GDP, earns foreign exchange through exports like cotton and rice, and supports rural incomes.

Skills: Suggest ways for agricultural improvements in Pakistan.

Answer: Give farmers better seed, balanced fertiliser and training in modern methods, and improve irrigation so less water is wasted.

Provide easy credit, crop insurance, cold storage and roads to markets so produce is not wasted and farmers get fair prices.

Support research, weather warnings and livestock health services, and teach farmers to protect soil and water.

Exercise — Chapter 4 (book pp. 57-58)

Q1 (i). Percentage of total World's cotton produced by Pakistan:

(a) 2%

(b) 3%

(c) 4%

(d) 5% ✓

Answer: (d) 5% — Pakistan is among the world's largest cotton producers, producing about five percent of world cotton.

Q1 (ii). The crop used for food and animal fodder:

(a) Wheat

(b) Maize ✓

(c) Gram

(d) Cotton

Answer: (b) Maize — maize grain is eaten by people and its stalks and leaves are fed to animals as fodder.

Q1 (iii). The contribution of agriculture in Pakistan's GDP is:

(a) 15%

(b) 20% ✓

(c) 25%

(d) 30%

Answer: (b) 20% — agriculture contributes about one-fifth of Pakistan's GDP, besides employing a large share of workers.

Q1 (iv). The method for growing many plants of different species in the same area:

(a) Crop Rotation

(b) Subsistence Agriculture

(c) Agroforestry

(d) Polyculture ✓

Answer: (d) Polyculture — polyculture means growing many different plant species together in the same area.

Q1 (v). Main crops of Pakistan are divided into major seasons:

(a) two ✓

(b) three

(c) four

(d) five

Answer: (a) two — Pakistan's main crops fall into the two seasons Rabi (winter) and Kharif (summer).

Q2 (i). What is meant by agriculture?

Answer: Agriculture is the cultivation of crops and the farming of cattle, also called husbandry.

It includes growing food crops, keeping animals, fishing and related farm work.

Most of the world's population depends on agriculture directly or indirectly for food and raw materials.

Q2 (ii). What is intercropping/mixed cropping?

Answer: Intercropping, or mixed cropping, means sowing more than two crops at the same time in the same field.

Different crops share the land, water and sunlight, and the mixture can reduce pests and improve soil use.

It differs from crop rotation, where crops are grown one after another in different seasons.

Q2 (iii). Write the name of five raw material obtained from farming.

Answer: Five raw materials obtained from farming are cotton, sugarcane, wheat, milk and wool.

Cotton feeds the textile industry, sugarcane gives sugar, and wheat and rice feed people.

Milk, hides and wool from cattle support the dairy, leather and woollen industries.

Q2 (iv). Name important crops of Pakistan.

Answer: The important crops of Pakistan are wheat, rice, cotton, sugarcane and maize.

Wheat and rice are staple foods, cotton supplies the textile industry, and sugarcane supplies sugar.

Maize is used for food and fodder, and vegetables such as potato, tomato and onion are also widely grown.

Q2 (v). Write the uses of cotton in three lines.

Answer: Cotton is the main raw material of the textile industry, spun into yarn and woven into cloth.

Its seeds yield cottonseed oil for cooking, and the leftover seed cake is used as animal feed.

Cotton therefore supports farming, industry, exports and millions of jobs in Pakistan.

Q3 (i). Explain the importance of agriculture in terms of Pakistan's economy.

Answer: Agriculture plays a vital role in Pakistan's economic growth, contributing about one-fifth of GDP.

It employs a large share of workers and provides food for the whole population.

It supplies raw materials such as cotton and sugarcane to industry, and its exports earn foreign exchange.

Q3 (ii). Describe different types of agriculture.

Answer: Subsistence agriculture is self-sufficiency farming, where farmers grow enough food crops to feed themselves and their families.

In intensive agriculture, crop growing and cattle breeding are done simultaneously on limited land.

In commercial agriculture, crops are produced on a large scale to earn profit, while traditional farming uses old methods practised for centuries.

Q3 (iii). Analyze the distribution of major crops in Pakistan.

Answer: Wheat is grown widely across Punjab and Sindh, and rice is concentrated in Punjab's rice belt and parts of Sindh.

Cotton is mainly a Punjab and Sindh Kharif crop, while sugarcane is grown in Punjab, Sindh and Khyber Pakhtunkhwa.

Maize is important in Punjab and Khyber Pakhtunkhwa, and vegetables of best quality are grown in many parts of Pakistan.

Q3 (iv). Write your point of view that Pakistan is an agricultural country.

Answer: Pakistan is an agricultural country because its fertile Indus Plain, irrigation system and two crop seasons suit farming.

A large share of its people work in farming, and agriculture feeds the nation and supplies industry.

Crops, livestock and fisheries together drive exports, jobs and the rural economy, proving agriculture's central place.

Q3 (v). Analyze the different methods of agriculture.

Answer: Traditional farming uses old tools, animal power and natural manure, and has been used for centuries.

Modern farming uses machinery, tube wells, chemical fertilisers and improved seeds for higher yields.

Other methods include subsistence, intensive and commercial farming, intercropping, and livestock farming alongside crops.

Q3 (vi). Enlist a few impacts of livestock (cattle) farming on the environment.

Answer: Cattle farming releases methane, a greenhouse gas, from animals and their waste, adding to climate change.

Overgrazing damages grasslands and soil, while animal waste can pollute water if not managed.

Large herds also need much fodder and water, putting pressure on land and water resources.

Q3 (vii). Discuss that raw material from farming and fishery help to generate revenues.

Answer: Farm raw materials such as cotton, sugarcane and wheat feed textile, sugar and flour industries that sell goods at home and abroad.

Fisheries make a significant contribution to the national economy through fish sales and exports.

Together they create jobs in farming, processing, transport and trade, and earn revenue and foreign exchange.

Critical Thinking Questions — Chapter 4 (book p. 58)

Critical Thinking: What is the most challenging issue facing agriculture today?

Answer: The most challenging issue facing agriculture today is water shortage, worsened by climate change.

Changing rainfall, hotter summers and shrinking river flows reduce irrigation water for crops.

Small landholdings, costly inputs and low productivity add to farmers' difficulties.

Critical Thinking: How can agriculture be a solution to a country's growth?

Answer: Agriculture drives growth by feeding the population, employing workers and supplying raw materials to industry.

Surplus crops and processed farm goods earn export income and foreign exchange.

Rising farm incomes also increase demand for goods and services, lifting the whole economy.

Critical Thinking: What is Organic Farming and why is it important?

Answer: Organic farming grows crops without chemical fertilisers and pesticides, using natural manure and biological pest control.

It is important because it protects soil health, keeps water and food free from chemical residues, and supports biodiversity.

It gives consumers healthier food and farmers a premium market, though yields may be lower.

Critical Thinking: What are the health risks, posed by the GMO Foods?

Answer: Concerns raised about GMO foods include possible allergic reactions to new proteins in modified crops.

Some people worry about antibiotic-resistance genes and unknown long-term effects on health.

These risks are debated among scientists, so labelling and safety testing of GMO foods are demanded.

Project For Students — Chapter 4 (book p. 58)

Project: Make a project on any one of the following topics: Problem and prospects of small scale fish farmers; Effect of rural-urban migration on agricultural production; Problems of deforestation; Problems affecting small scale farmers.

Answer: Project task (to be made, not written): choose any one of the four topics listed in the question.

Collect facts, pictures and examples, and present causes, effects and possible solutions.

Present the project to the class and explain what the topic means for farming and for Pakistan.

Chapter 5 — Climate Change

Skills — Chapter 5 (book p. 61)

Skills: Use weather charts and climatic maps to identify main areas where major climatic changes are occurring.

Answer: Compare temperature and rainfall charts for the same place over many years to spot warming, drying or heavier rain.

On climatic maps mark areas with rising temperatures, melting glaciers, longer droughts and more frequent floods.

In Pakistan this points to the northern glacier areas, the flood-prone Indus Plain and the drought-hit southern regions.

Skills — Chapter 5 (book p. 63)

Skills: Investigate how human activity is causing climate change.

Answer: Burning coal, oil and gas for power, transport and industry releases carbon dioxide, the main greenhouse gas.

Cutting forests removes trees that absorb carbon dioxide, while farming, livestock and waste add methane and other gases.

These gases trap heat near the Earth, so the atmosphere warms and the climate changes.

Skills: Investigate the effects of global warming on polar regions.

Answer: Warming melts sea ice, glaciers and ice sheets in the Arctic and Antarctic, and the meltwater raises sea levels.

Polar animals such as polar bears and seals lose their ice habitat and food, and thawing permafrost releases more greenhouse gases.

Less white ice also means less sunlight is reflected, so the poles warm even faster.

Skills: Analyze the effect of climatic change on people, lifestyle, and economy.

Answer: Heatwaves, floods and droughts damage homes, harm health and force people to move from affected areas.

Crops fail, water becomes scarce and food prices rise, changing how families live and work.

Repairing flood damage, losing harvests and paying for health care cost the economy heavily, especially in poor countries.

Skills — Chapter 5 (book p. 64)

Skills: Analyze the effects of climate change on the Earth.

Answer: Climate change raises average temperatures, melts glaciers and ice caps, and raises sea levels, flooding coastal areas.

It makes extreme weather more common: heavier floods, longer droughts, stronger storms and more wildfires.

Ecosystems are disturbed as species lose habitats, coral reefs bleach and deserts spread into fertile land.

Skills: Evaluate the health and environmental effects of smog in Pakistan.

Answer: Smog harms health: it irritates eyes and throat, worsens asthma and breathing diseases, and is dangerous for children and the elderly.

It cuts visibility, disrupts traffic and flights, damages crops and buildings, and keeps sunlight from reaching the ground.

In Pakistan winter smog in cities like Lahore comes from vehicles, industry, brick kilns and crop burning, and needs cleaner fuel and controls.

Skills: Investigate and suggest ways to reduce the greenhouse effect.

Answer: Cut greenhouse gases by using solar, wind and hydropower instead of coal, oil and gas, and by saving electricity.

Plant trees, protect forests, use public transport, and follow the 3Rs — reduce, reuse and recycle.

Improve energy efficiency in homes and factories, and reduce waste, so less heat is trapped in the atmosphere.

Exercise — Chapter 5 (book pp. 70-71)

Q1 (i). The gas responsible for rise in the Earth's temperature:

(a) sulphur dioxide

(b) hydrogen

(c) nitric oxide

(d) carbon dioxide ✓

Answer: (d) carbon dioxide — carbon dioxide is the main greenhouse gas

trapping heat and raising Earth's temperature.

Q1 (ii). Hydro chloro fluoro carbons (HCFCs) is a:

(a) gas ✓

(b) liquid

(c) solid

(d) power plant

Answer: (a) gas — HCFCs are gases used in refrigeration and air conditioning that

harm the ozone layer.

Q1 (iii). Example of green house gas:

(a) helium

(b) hydrogen

(c) carbon dioxide ✓

(d) nitrogen

Answer: (c) carbon dioxide — carbon dioxide, methane and water vapour are greenhouse gases; helium, hydrogen and nitrogen are not.

Q1 (iv). Earth's warmest month was recorded in:

(a) July 2019 ✓

(b) July 2020

(b) July 2021

(d) July 2022

Answer: (a) July 2019 — at the time of this textbook, July 2019 stood as the warmest month ever recorded on Earth. (Note: the book misprints option (b) twice.)

Q1 (v). In 2009, an International Conference about global climate change was held in:

(a) Seoul

(b) London

(c) Tokyo

(d) Copenhagen ✓

Answer: (d) Copenhagen — the 2009 UN climate conference (COP15) was held in Copenhagen, Denmark.

Q2 (i). Differentiate between climate and weather.

Answer: Weather is the short-term state of the atmosphere — temperature, rain and wind on a particular day.

Climate is the long-term average pattern of weather in a region, usually over about thirty years.

Weather changes daily, while climate changes only slowly over decades.

Q2 (ii). What is Global Warming?

Answer: Global warming is the gradual rise in the Earth's average temperature.

It is caused mainly by greenhouse gases from burning fossil fuels, which trap extra heat in the atmosphere.

It leads to melting glaciers, rising seas and more extreme weather.

Q2 (iii). Which gas absorbs the heat coming from the Earth?

Answer: Carbon dioxide is the main gas that absorbs heat radiated from the Earth's surface.

Other greenhouse gases such as methane and water vapour also absorb outgoing heat.

By absorbing this heat, they warm the atmosphere instead of letting it escape to space.

Q2 (iv). Write a few benefits of Ozone gas.

Answer: Ozone in the stratosphere absorbs most of the Sun's harmful ultraviolet rays before they reach Earth.

This protects people from skin cancer and eye damage, and protects crops and animals.

Without the ozone layer, life on Earth would face far more harmful radiation.

Q2 (v). Define the term greenhouse effect.

Answer: The greenhouse effect is the warming of the Earth when greenhouse gases trap heat in the atmosphere.

Sunlight warms the Earth's surface, which radiates heat upward, and greenhouse gases absorb part of that heat.

This trapped heat keeps the Earth warm enough for life, but extra gases strengthen the effect and cause warming.

Q3 (i). Analyze the effect of climatic change on the Earth.

Answer: Climate change raises average temperatures, melts glaciers and polar ice, and raises sea levels.

It makes rainfall irregular, causing droughts in some regions and floods in others.

Heatwaves, storms and wildfires grow more frequent, harming farming, health, wildlife and coastal settlements.

Q3 (ii). Describe some climatic change related events in the world.

Answer: The world has seen record heatwaves, melting Arctic ice and rising sea levels in recent decades.

Severe floods, such as Pakistan's floods in 2022, and hurricanes, droughts and wildfires have become more common.

Shrinking glaciers and shifting rainfall patterns show climate change is already affecting every continent.

Q3 (iii). What changes may be caused by the greenhouse effect? Describe.

Answer: A stronger greenhouse effect raises global temperatures and melts ice caps and glaciers.

Sea levels rise as ice melts and warm water expands, threatening coastal cities and deltas.

Rainfall patterns shift, growing seasons change, and extreme weather such as heatwaves, floods and droughts increases.

Q3 (iv). Do you know that Ozone layer is very important? Explain.

Answer: Yes, the ozone layer is very important because it acts as a shield against the Sun's ultraviolet rays.

It absorbs most UV radiation, protecting human health, crops, animals and marine life.

If the ozone layer thins, skin cancer, eye damage and harm to plants and plankton all increase.

Q3 (v). Write a note on 3R's.

Answer: The 3R's are Reduce, Reuse and Recycle — three steps for cutting waste and protecting the environment.

Reduce means using less, for example refusing plastic bags and saving paper and energy.

Reuse means using items again, such as bottles and bags, and Recycle means turning waste material into new products.

Critical Thinking Questions — Chapter 5 (book p. 71)

Critical Thinking: Suggest some ways to reduce plastic use in daily life?

Answer: Carry cloth bags for shopping, use reusable water bottles, and refuse single-use plastic straws and cutlery.

Choose products with less packaging, reuse plastic containers, and sort plastic waste for recycling.

Small daily choices by many people greatly reduce the plastic reaching landfills and rivers.

Critical Thinking: Why do we care about Ozone depletion?

Answer: We care about ozone depletion because a thinner ozone layer lets more harmful ultraviolet rays reach Earth.

Extra UV radiation increases skin cancer and eye cataracts, damages crops, and harms plankton that support ocean food chains.

Protecting ozone therefore protects human health, farming and the whole environment.

Critical Thinking: What can we do to stop green house effect?

Answer: We can slow the greenhouse effect by burning fewer fossil fuels and using solar, wind and hydropower instead.

Saving electricity, planting trees, using public transport and reducing waste all cut greenhouse gas emissions.

International agreements and cleaner industries are also needed for a lasting solution.

Critical Thinking: How can we use the renewable energy?

Answer: We can use renewable energy by installing solar panels on roofs to make electricity and heat water.

Wind turbines generate electricity from wind, hydropower plants use flowing water, and biogas plants use organic waste.

Renewable energy is clean, never runs out, and reduces pollution and climate change.

Project For Students — Chapter 5 (book p. 71)

Project: Project 3Rs is all about 'Reuse, Reduce and Recycle'. It is to create awareness among children, parents and the community about the importance of reuse, reduce and recycle.

Answer: Project task (to be made, not written): run a 3Rs awareness project in your school and neighbourhood.

Show examples of reducing waste, reusing items and recycling materials, with charts and collected items.

Explain to children, parents and the community why Reduce, Reuse and Recycle protect the environment.

Chapter 6 — Minerals and Powers

Skills — Chapter 6 (book p. 80)

Skills: Locate areas of Pakistan that are rich in mineral resources on maps.

Answer: On a map of Pakistan mark the mineral-rich areas: salt in Khewra and the Salt Range, coal in Thar, Lakhra and Dandot.

Mark copper and gold at Saindak and Reko Diq in Balochistan, chromite in Muslim Bagh and Chaghi, and iron ore at Kalabagh.

Use symbols from the book's minerals map and a key naming each mineral.

Skills: Categorize the minerals found in Pakistan according to their use and importance.

Answer: Fuel minerals — coal, oil and natural gas — are used to produce energy and run industry and transport.

Metallic minerals — iron ore, copper, chromite and gold — are used in steel, cables, machinery and jewellery.

Non-metallic minerals — rock salt, gypsum, marble and limestone — are used in cooking, cement, fertiliser and building.

Skills: Analyze factors that make gemstone mining difficult in Pakistan.

Answer: Gemstone mines are often in remote, mountainous areas with poor roads, so machinery and workers are hard to bring in.

Mining needs skilled workers, safety equipment and money, which small miners often lack, and unplanned digging wastes precious stones.

Lack of cutting, polishing and marketing facilities means raw stones are sold cheaply instead of earning full value.

Skills: Analyze the reasons that hamper the yield of gemstone mining in Pakistan.

Answer: Old, unscientific methods break stones and recover only a small part of the deposit, lowering the yield.

There is little geological survey work, so miners dig without knowing where the best deposits lie.

Weak safety rules, shortage of trained workers and poor transport further reduce how much good-quality stone is mined.

Skills: Suggest practical ways of getting high-income revenue from Pakistan's gemstones resources.

Answer: Cut, polish and set gemstones in Pakistan instead of exporting rough stones, so the value stays in the country.

Train workers in modern mining and gem cutting, and hold exhibitions and online sales to reach world buyers.

Give miners fair prices, proper licences and branding for Pakistani stones like emerald, ruby and topaz.

Skills: Investigate and suggest measures for improving the mining industry.

Answer: Survey deposits properly, use modern and safe machinery, and train miners in safe methods.

Build roads and power to mining areas, give miners credit and fair prices, and enforce safety and environmental rules.

Set up processing plants near mines so minerals are refined in Pakistan and create skilled jobs.

Skills: Suggest measures to ensure the safety and health of the coal miners in Pakistan.

Answer: Give miners helmets, lamps, masks and strong supports in tunnels, and check roofs and gas levels before every shift.

Provide ventilation, clean drinking water, first aid and regular health check-ups for dust-related lung disease.

Train miners in rescue and safety rules, and make owners follow the law on working hours and compensation.

Skills — Chapter 6 (book p. 84)

Skills: Draw and label the procedures of long-wall and surface mining.

Answer: Long-wall mining (underground): a long coal face is cut by machine, the coal falls onto a conveyor, and the roof is supported as the face advances.

Surface mining (open-pit/strip): the top soil and rock are removed to expose the mineral, which is then dug out in steps or benches.

Label the coal face, conveyor, roof supports and waste rock in the underground diagram, and the benches, mineral layer and overburden in the surface diagram.

Skills: Analyze coal usage in Pakistan.

Answer: In Pakistan coal is used for thermal power production, brick kilns and domestic needs.

It is a cheap fuel, but its use is limited by pollution and by the availability of better fuels.

Coal mining also provides jobs in mining regions, while its smoke and ash harm air quality.

Skills: Evaluate the usefulness of coal against the risk factors associated with it.

Answer: Coal is useful because it is cheap, abundant and can generate electricity and fire brick kilns.

Its risks include mining accidents, roof collapse, gas explosions, flooding and lung disease from coal dust.

Burning coal also pollutes air and adds greenhouse gases, so its benefits must be weighed against health and environmental costs.

Skills: Identify the importance of alternate sources of energy.

Answer: Alternate sources such as solar, wind and hydropower are important because they never run out.

They produce no smoke and no greenhouse gases, giving clean air and water and cheaper electricity over time.

They also create jobs and help reverse climate change, as listed in the chapter's benefits of renewable energy.

Skills: Explore reasons for the scarcity of electricity and gas in Pakistan in recent years.

Answer: Electricity and gas are scarce in Pakistan because demand has grown faster than new power plants and gas fields.

Dependence on imported fuels, circular debt and old transmission lines add to shortages and load shedding.

Limited investment, losses and rapid urban growth have worsened the gap between supply and demand.

Exercise — Chapter 6 (book pp. 84-85)

Q1 (i). Reserves of mineral in Pakistan are approximately more than 220 million tons:

(a) Coal

(b) Salt ✓

(c) Gypsum

(d) Iron

Answer: (b) Salt — Pakistan's rock salt reserves, centred on the Khewra Salt Mines, are estimated at more than 220 million tons.

Q1 (ii). Extraction of valuable minerals and other material from the Earth is called:

(a) Mining ✓

(b) Geography

(c) Geology

(d) Geodesy

Answer: (a) Mining — the chapter's summary states extraction of valuable minerals and other materials from the Earth is called mining.

Q1 (iii). In Pakistan, natural gas was discovered in:

(a) Sindh

(b) Punjab

(c) Khyber Pakhtunkhwa

(d) Balochistan ✓

Answer: (d) Balochistan — the chapter's summary states natural gas was discovered at Sui in Sibi, Balochistan.

Q1 (iv). Production of iron ore in Pakistan was started in:

(a) 1953 CE

(b) 1955 CE

(c) 1957 CE ✓

(d) 1959 CE

Answer: (c) 1957 CE — the chapter's summary states production of iron ore in Pakistan was started in 1957 CE.

Q1 (v). Energy obtained from sunlight is called:

(a) Wolar power

(b) Solar power ✓

(c) Hydro power

(d) Nuclear power

Answer: (b) Solar power — the chapter's summary states energy extracted from sunlight is called solar power. (The book prints option (a) as 'Wolar power'.)

Q2 (i). What is the difference between metallic and non-metallic minerals?

Answer: Metallic minerals contain metals, look shiny, and can be melted and shaped; examples are iron ore and copper.

Non-metallic minerals contain no metals and are usually dull and brittle; examples are salt, gypsum and coal.

Metallic minerals are used for metals and machinery, while non-metallic minerals serve chemical, building and fuel uses.

Q2 (ii). What are the uses of Coal?

Answer: Coal is used for thermal power production, generating electricity in power plants.

It fires brick kilns and cement works, and meets domestic needs for heating and cooking.

Coal is therefore an important, though polluting, fuel for industry and homes in Pakistan.

Q2 (iii). How wind-electricity is generated?

Answer: Wind-electricity is generated when moving air turns the large blades of a wind turbine.

The spinning blades turn a generator inside the turbine, which converts motion into electricity.

The electricity is then sent through wires to homes, farms and industries.

Q2 (iv). Define minerals and give examples.

Answer: Minerals are naturally occurring solid substances found in the Earth, formed by geological processes.

They have a definite composition and are dug out by mining.

Examples from the chapter are coal, salt, gypsum, chromite and iron ore.

Q2 (v). What are the major products of mineral oil?

Answer: The major products of mineral oil include gasoline, kerosene oil, wax and coal tar.

Gasoline fuels vehicles, kerosene is used for lighting and cooking, and wax is used in candles and polish.

Other products include diesel, lubricants and raw materials for the chemical industry.

Q3 (i). Describe the minerals found in Pakistan.

Answer: Pakistan has coal, used for power and brick kilns, and rock salt from the Khewra mines.

Gypsum is used as raw material in phosphate fertiliser, and chromite reserves number more than 25 deposits.

Iron ore, natural gas from Sui, mineral oil, limestone and other minerals are also found across the country.

Q3 (ii). Explain industries related to minerals and power in Pakistan.

Answer: Mineral industries include coal mining for thermal power, salt mining at Khewra, and cement works using limestone and gypsum.

Fertiliser plants use gypsum, steel works use iron ore, and oil refineries process mineral oil.

Power industries use coal, gas, oil and hydropower: Ghazi Barotha and Mangla are major hydropower projects.

Q3 (iii). Discuss the use of fossil fuels (coal, oil, and gas) to generate power.

Answer: Fossil fuels are burned to produce heat, which makes steam that drives turbines connected to generators.

Coal fuels thermal power plants, oil runs engines and generators, and natural gas fires efficient power stations.

They give reliable power but release smoke and greenhouse gases, so cleaner alternatives are growing.

Q3 (iv). Describe the types of mining.

Answer: The two main types of mining are surface mining and underground mining.

In surface mining, the top layers of soil and rock are removed to expose minerals near the surface.

In underground mining, shafts and tunnels reach deep deposits, and methods such as long-wall mining extract coal along a long face.

Q3 (v). Analyze the risk factors associated with coal mining.

Answer: Coal miners face roof collapse, gas explosions, flooding and accidents with heavy machinery underground.

Coal dust damages lungs over time, causing breathing diseases among miners.

Mining also harms the environment through land damage, waste heaps and polluted water near mines.

Q3 (vi). Explain alternate sources of energy (solar, wind, and water) to generate power.

Answer: Solar power uses panels that convert sunlight directly into electricity, with no fuel and no smoke.

Wind power uses turbines whose blades turn generators as wind blows.

Water power, or hydropower, uses dams and flowing rivers: falling water turns turbines, and projects like Ghazi Barotha and Mangla supply cheap electricity.

Q3 (vii). Highlight the importance of alternative energy sources for the environment.

Answer: Alternative energy sources give a fuel supply that never runs out and produce zero carbon emissions.

They give clean air and water, no greenhouse gas emissions and no pollution, and a cheaper form of electricity.

They create new jobs and help reverse climate change, making them vital for the environment's future.

Critical Thinking Questions — Chapter 6 (book p. 86)

Critical Thinking: Explore some common challenges faced by the mining industry.

Answer: The mining industry faces safety dangers such as collapse, explosions and flooding in underground mines.

It faces high costs, falling ore quality as easy deposits run out, and remote, difficult locations.

Environmental rules, water use, waste disposal and price swings in mineral markets add further challenges.

Critical Thinking: Explore the impacts of mining on the environment.

Answer: Mining removes vegetation and topsoil, damages land and leaves waste heaps and open pits.

It can pollute air with dust and water with chemicals and sediments from mines.

Habitats are destroyed, and abandoned mines may keep leaking pollutants unless the land is restored.

Critical Thinking: Why solar energy is important for future?

Answer: Solar energy is important for the future because sunlight is free, abundant and never runs out.

It produces electricity without smoke, carbon emissions or fuel imports, and panels work even in remote areas.

As panel costs fall, solar power can meet growing demand while protecting the climate.

Critical Thinking: What are pros and cons of wind energy?

Answer: Pros of wind energy: it is renewable, clean, produces no emissions, and costs little once turbines are built.

Cons: wind is intermittent, so output varies; turbines need open land, can be noisy, and may affect birds.

Despite these limits, wind power remains a valuable clean source alongside solar and hydropower.

Project For Students — Chapter 6 (book p. 87)

Project: Make a model of a mine by showing the process of Mining.

Answer: Project task (to be made, not written): build a model of a mine showing shafts, tunnels, the mineral layer and transport of ore.

Label the mining process from digging to bringing the mineral to the surface.

Present the model to the class and explain the steps and safety measures in mining.